Mihai Dincă

Date of birth: 31/08/2003 | **Place of birth:** Drăgășani, Romania |

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ABOUT ME

Third-year student at the Faculty of Electronics, Telecommunications and Information Technology, specializing in Applied Electronics. Motivated to continuously learn, gain experience, and get involved in hands-on projects that contribute to my professional development.

EDUCATION AND TRAINING

2023 – CURRENT Cluj-Napoca, Romania

STUDENT Technical University of Cluj-Napoca

Website https://etti.utcluj.ro/ | **Field of study** Applied Electronics

PROJECTS

PWM Signal Generator Design and Simulation

Objective: Design and simulation of a Pulse-Width Modulation (PWM) signal generator with adjustable amplitude and duty cycle.

Implementation:

- Circuit implemented using three operational amplifiers: 2 comparators and integrator
- Charge/discharge path separation circuit for the capacitor
- Voltage divider used for amplitude and duty cycle adjustment
- Circuit design and simulation performed in OrcadX

Skills gained:

- Circuit design and simulation using OrcadX
- Understanding the operating principles of PWM signals
- Analysis and use of Schmitt trigger circuits

Low-Amplitude Signal Processing Chain

Objective: Design of a signal processing chain for low-amplitude signals.

Implementation:

- Circuit structured into four stages: amplifier, Tow-Thomas band-pass filter, programmable gain amplifier (PGA), and full-wave rectifier
- Software implementation: signal integrity testing using OP, Transient, and AC Sweep analyses in LTspice
- Hardware implementation: circuit built on a breadboard and performance validated using measurement instruments

Skills gained:

- Analog circuit design
- Signal analysis in both time and frequency domains
- Hardware implementation and use of measurement tools (Scoppy)

Digital Clock on FPGA (VHDL)

Objective: Design and implementation of a digital clock on an FPGA board with time and minute setting functions, stopwatch functionality, and display on a 7-segment display using VHDL.
Implementation:

- Designed a Finite State Machine (FSM) to control operating modes: time display, hour setting, minute setting, and stopwatch
- Implemented a frequency divider to generate timing signals
- Developed increment logic for hours, minutes, and seconds in 24-hour format
- Managed the 7-segment display for time visualization
- Implemented a stopwatch function with start/stop control and display of seconds and subdivisions

Skills gained:

- Digital system design and implementation using VHDL
- Clock signal management and frequency division in digital systems
- Implementation and testing of FPGA projects on platforms such as the Basys 3 FPGA Board

LANGUAGE SKILLS

Mother tongue(s): **ROMANIAN**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
ENGLISH	B2	B2	B1	B2	B2

Levels: A1 and A2: Basic user; B1 and B2: Independent user; C1 and C2: Proficient user

HOBBIES AND INTERESTS

Technical Interests

- Analog and digital circuit design
- Signal processing and analysis
- Electronic circuit simulation (OrcadX, LTspice)
- Hardware prototyping and testing

COMPETENȚE TEHNICE

Software Skills

- **VHDL:** Basic level
- **Assembly:** Intermediate level
- **C/C++:** Intermediate level
- **Java:** Intermediate level
- **Python:** Intermediate level
- **Scoppy & WaveForms:** Signal simulation, acquisition, and analysis
- **Microsoft Office:** Word, PowerPoint

Soft Skills

- Strong teamwork skills
- Good organization and ability to meet deadlines
- Analytical thinking and problem-solving abilities

DRIVING LICENCE

Driving Licence: B